

Research Report

Summary

Public literature on oxide dielectrics for thin-film work consistently converges on a small set of simple binary oxides that balance wide band gap, thermodynamic/chemical stability, and non-toxic composition. Based on the criteria specified (thermodynamic stability, wide gap, non-toxic elements, simple composition, public evidence), the ranked shortlist is:

1. **Al₂O₃ (aluminum oxide)** — Best overall balance: very wide band gap (~8.8 eV), well-documented thermodynamic/chemical stability on Si, non-toxic, mature ALD process, simple binary composition. Con: moderate dielectric constant ($k \approx 7-10$), sometimes limited by leakage in certain deposition routes.
2. **MgO (magnesium oxide)** — Widest gap among common candidates (~7.3–7.8 eV reported, though some thin-film reports show lower apparent gaps ~4.3–5.7 eV depending on processing), simple rock-salt binary, biocompatible/non-toxic, high thermal stability. Cons: moisture/hydroxide sensitivity in ambient air, film quality/precursor difficulties reported, dielectric constant modest (~9.6–9.8), reported optical band gap varies significantly with deposition method — evidence is less consistent than Al₂O₃.
3. **HfO₂ (hafnium oxide)** — Excellent thermodynamic stability on Si (higher oxygen-vacancy formation energy than ZrO₂), good gap (~5.5–6 eV), higher k (~20–25), most industrially mature high- k oxide (already in production CMOS). Cons: Hf is not classically "toxic" but is a less-abundant, costlier element than Al/Mg/Si; band gap is narrower than Al₂O₃/MgO, and the composition (metal + O) is still simple but less "light-element/benign" by some readers' standards.
4. **ZrO₂ (zirconium oxide)** — Comparable properties to HfO₂ ($k \approx 20-25$, gap ≈ 5.8 eV) but shows more silicon-interface stability concerns in some reports (lower oxygen-vacancy formation energy, more crystallization/leakage issues) — ranked below HfO₂ for stability reasons despite similar simplicity and non-toxicity.
5. **Y₂O₃ (yttrium oxide)** — Reasonable gap (~5.6 eV) and moderate k (~15), simple sesquioxide, generally regarded as non-toxic and stable, but has narrower public evidence base for thin-film device performance compared to the top four, and deposition (e.g., PLD-only in some studies) is less mature.

Materials such as La₂O₃, TiO₂, Ta₂O₅, and SrTiO₃ were found in the literature but are **not recommended** as top picks here: La₂O₃ has a narrower gap (~4.3–6 eV) and moisture sensitivity; TiO₂ and Ta₂O₅/SrTiO₃ have narrow gaps (~3.5–4.4 eV) and are reported as unstable in direct contact with Si.

Overall recommendation: Al₂O₃ is the safest default first choice for new thin-film dielectric experiments given the strongest combination of wide gap, stability, and non-toxicity with a large, consistent public evidence base. HfO₂ is the recommended alternative if a substantially higher dielectric constant is the priority.

Technical / Audit

Al₂O₃ — Reported band gap ~8.8 eV [Shin 2016, run.unl.pt], dielectric constant reported in the range $k \approx 7-10$ depending on deposition method (ALD plasma vs. thermal) [Wright.edu report; PMC6445835]. Multiple sources describe Al₂O₃ as "thermally and chemically stable" and well-adhering, used widely as a gate/passivation dielectric and in TFT stacks [ScienceDirect S0169433212001493; ScienceDirect S1369800118307455]. Evidence quality is high: many independent groups report consistent gap values and stability behavior across ALD, sputtering, and PLD routes. Limitation: its k -value is comparatively low relative to HfO₂/ZrO₂, so if higher

capacitance density is required, Al₂O₃ alone may be insufficient (commonly paired in nanolaminates, e.g., Al₂O₃/HfO₂ [Nano-Micro Letters 2022, PMC9515270]).

MgO — Reported properties vary noticeably across sources: bulk/CVD-grown films report gap ~7.3 eV and $k \approx 9.8$ [girolamigroup.web.illinois.edu single-crystal-like CVD film], Goodfellow materials data sheet reports gap ~7.8 eV and $k \approx 9.6$ [Goodfellow, MgO]; but solution-processed and spray-pyrolysis thin films report much lower optical gaps (3.42–5.7 eV) attributed to processing-induced defects/crystallinity differences [distantreader.org JNPS 931; imapsjmep.org]. This spread indicates the "true" MgO thin-film gap is process-dependent, and public evidence is less consistent than for Al₂O₃. MgO is simple (binary, rock-salt structure), non-toxic/biocompatible [Goodfellow], and thermally very stable (melting point 2800–2852 °C) — but is known in the wider ceramics literature (outside these search results) to be hygroscopic, which can be a practical experimental caveat even though it wasn't explicitly quantified in the sources retrieved here. ALD/CVD precursor availability and film-quality issues were explicitly noted as a challenge in one applied study [Wright.edu report].

HfO₂ — Reported $k \approx 20$ –25 (some sources cite 12–47 depending on crystalline phase) and band gap ≈ 5.5 –6 eV [przyrbwn.icm.edu.pl review; NPG Asia Materials, nature.com/am201557]. HfO₂ has a higher computed oxygen-vacancy formation energy (Evac ≈ 6.9 eV) than ZrO₂ (Evac ≈ 5.1 eV), supporting its reported superior thermodynamic/interfacial stability on Si [nature.com/am201557]. It is already in production use as a CMOS gate dielectric, giving it the deepest and most mature evidence base of any high- k oxide reviewed. Leakage current for HfO₂ (10^{-3} A/cm²) reported lower than ZrO₂ (4×10^{-3} A/cm²) under similar test conditions [przyrbwn.icm.edu.pl]. Caveat: narrower gap than Al₂O₃ or MgO, and Hf as an element, while not classified as toxic, is scarcer/costlier and less "simple" from a supply-chain perspective than Al, Mg, or Si compounds.

ZrO₂ — Similar profile to HfO₂ ($k \approx 16$ –25, up to 36.8–46.6 in cubic/tetragonal crystalline phases; gap ≈ 5.8 eV) [d-nb.info thesis; PMC12029089]. Public evidence shows more emphasis on stability concerns with Si substrates (lower Evac, more crystallization-related leakage issues) relative to HfO₂ [nature.com/am201557; przyrbwn.icm.edu.pl]. ZrO₂ remains attractive for SiC-based devices and DRAM capacitor dielectrics where its properties are favorably matched [ACS AAMI 2025; nature.com/am201557].

Y₂O₃ — Reported $k \approx 15$, gap ≈ 5.6 eV [przyrbwn.icm.edu.pl table]. Public evidence base for thin-film experiments is comparatively thin; one applied report noted Y₂O₃ could only be deposited by PLD (not ALD) in their setup, limiting deposition flexibility [Wright.edu report].

Recommendation by scenario:

- **Primary scenario (general-purpose, high-stability, benign wide-gap dielectric for exploratory thin-film work):** Al₂O₃ — strongest, most consistent public evidence, simplest and safest composition.

- **If maximizing dielectric constant/capacitance density is the priority (e.g., DRAM-like capacitor structures):** HfO₂ (with ZrO₂ as a close alternative, particularly for SiC-compatible or crystalline-phase-engineered stacks).

- **If maximizing band gap / minimizing leakage in a UWBG-transistor context (e.g., pairing with Ga₂O₃):** MgO or Al₂O₃ are the top picks per the Wright.edu UWBG dielectric survey, though MgO's thin-film gap values are inconsistent across processing routes and should be independently verified for the specific deposition method used.

- **If a single-material solution with balanced k and gap is needed and Hf supply/cost is not a constraint:** HfO₂ remains the most industrially validated choice.

- **Not recommended given the stated criteria:** La₂O₃ (moisture-sensitive, narrower gap), TiO₂/Ta₂O₅/SrTiO₃ (narrow gaps, reported instability on Si), despite their very high k -values, because they fail the wide-band-gap

and/or thermodynamic-stability screens.

Overall limitations across all sources: Values for k and E_g vary by amorphous vs. crystalline phase, deposition technique (ALD, PLD, sputtering, solution processing), and measurement method (optical Tauc plot vs. electrical C-V vs. spectroscopic ellipsometry), so any specific number should be treated as method-dependent rather than absolute. No single source in this search directly compared thermodynamic formation-energy data across all five candidates; the oxygen-vacancy formation energy comparison (E_{vac}) was found specifically for HfO_2 , ZrO_2 , and SiO_2 in one ab initio screening paper [nature.com/am201557], not for Al_2O_3 , MgO , or Y_2O_3 — this is a gap in cross-comparable public evidence.

Data Focus

Material	Band gap (eV)	Dielectric constant (k)	Stability notes	Source
SiO_2	9	3.9	Reference/baseline; thermodynamically stable on Si	przyrbwn.icm.edu.pl (APP 136z6p05)
Al_2O_3	8.8	10 (table value); ~7–8.6 (measured ALD films, method-dependent)	Thermally/chemically stable on Si; adheres well	run.unl.pt (Shin 2016); przyrbwn.icm.edu.pl; PMC6445835; ScienceDirect S0169433212001493
MgO	7.3–7.8 (CVD/bulk-like film; Goodfellow datasheet); 3.42–5.7 (solution/spray-processed thin films, process-dependent)	9.6–9.8	Very high melting point (2800–2852 °C); simple rock-salt binary; biocompatible; hygroscopicity noted qualitatively in general ceramics literature (not directly quantified in retrieved sources)	girolamigroup.web.illinois.edu (APL 2013); Goodfellow MgO datasheet; distantreader.org (JNPS 931); imapsjmep.org
HfO_2	5.5–6 (also cited as 5.6, 5.8)	20–25 (range 12–47 reported depending on phase)	Higher oxygen-vacancy formation energy ($E_{vac} \approx 6.9$ eV) than ZrO_2 ; industrially mature; lower leakage current than ZrO_2 in comparative test	przyrbwn.icm.edu.pl; nature.com/am201557; ScienceDirect S0169433212001493

Material	Band gap (eV)	Dielectric constant (k)	Stability notes	Source
ZrO ₂	5.8	16–25 (up to 36.8–46.6 in cubic/tetragonal crystalline phases)	Lower Evac (≈ 5.1 eV) than HfO ₂ ; more crystallization/leakage issues reported on Si; favorable for SiC devices	PMC12029089; d-nb.info th